WORK REPORT

Table 1: Runtime and memory for QDFT ActiveSpace(2e,4o)

Molecule	Formula	DFT (s)	Embedding (s)	VQE (s)	Total (s)	Peak RSS (MB)
Benzene	C_6H_6	17.21	5.12	32.57	54.90	858.30
Naphthalene	$C_{10}H_8$	32.66	33.02	54.51	120.19	1956.70
Anthracene	$C_{14}H_{10}$	44.91	129.68	110.64	285.23	5326.03
Pyrene	$C_{16}H_{10}$	79.27	204.60	153.07	436.94	4771.27

Table 6: QDFT ActiveSpace(2e,4o) Energy Results

Molecule	Formula	Electronic Ground State Energy (Eh)	Nuclear Repulsion Energy (Eh)	Total Ground State Energy (Eh)
Benzene	C_6H_6	-434.1148	203.1468	-230.9680
Naphthalene	$C_{10}H_8$	-840.8874	457.1060	-383.7814
Anthracene	$C_{14}H_{10}$	-1301.6774	765.0882	-536.5893
Pyrene	$C_{16}H_{10}$	-1577.6300	965.1896	-612.4403

Table 2: Runtime and memory for QDFT ActiveSpace(2e,6o)

Molecule	Formula	DFT (s)	Embedding (s)	VQE (s)	Total (s)	Peak RSS (MB)
Benzene	C_6H_6	16.94	5.17	178.24	205.07	859.14
Naphthalene	$C_{10}H_8$	32.84	34.43	198.83	275.11	2135.02
Anthracene	$C_{14}H_{10}$	45.55	125.78	258.49	448.88	5496.84
Pyrene	$C_{16}H_{10}$	78.87	209.93	321.33	645.47	5032.90

Table 7: QDFT ActiveSpace(2e,6o) Energy Results

Molecule	Formula	Electronic Ground State Energy (Eh)	Nuclear Repulsion Energy (Eh)	Total Ground State Energy (Eh)
Benzene	C_6H_6	-434.1145	203.1468	-230.9677
Naphthalene	$C_{10}H_8$	-840.8870	457.1060	-383.7810
Anthracene	$C_{14}H_{10}$	-1301.6779	765.0882	-536.5897
Pyrene	$C_{16}H_{10}$	-1577.6306	965.1896	-612.4409

Table 3: Runtime and memory for QDFT ActiveSpace(4e,6o)

Molecule	Formula	DFT (s)	Embedding (s)	VQE (s)	Total (s)	Peak RSS (MB)
Benzene	C_6H_6	17.57	5.34	1075.21	1102.80	1095.72
Naphthalene	$C_{10}H_8$	32.64	33.81	1106.45	1181.92	2375.45
Anthracene	$C_{14}H_{10}$	44.77	125.65	1182.46	1374.65	5670.64
Pyrene	$C_{16}H_{10}$	81.18	207.39	1234.49	1558.57	5114.06

Table 8: QDFT ActiveSpace(4e,6o) Energy Results

Molecule	Formula	Electronic Ground State Energy (Eh)	Nuclear Repulsion Energy (Eh)	Total Ground State Energy (Eh)
Benzene	C_6H_6	-434.1365	203.1468	-230.9897
Naphthalene	$C_{10}H_8$	-840.9040	457.1060	-383.7980
Anthracene	$C_{14}H_{10}$	-1301.6916	765.0882	-536.6034
Pyrene	$C_{16}H_{10}$	-1577.6429	965.1896	-612.4533

Table 4: Runtime and memory for QDFT ActiveSpace(6e,6o)

Molecule	Formula	DFT (s)	Embedding (s)	VQE (s)	Total (s)	Peak RSS (MB)
Benzene	C_6H_6	17.15	5.31	1742.78	1769.97	1094.59
Naphthalene	$C_{10}H_8$	32.80	33.98	1753.81	1829.63	2353.64
Anthracene	$C_{14}H_{10}$	43.45	125.28	1789.03	1975.83	5746.43
Pyrene	$C_{16}H_{10}$	77.14	210.28	1965.36	2285.51	5346.21

Table 9: QDFT ActiveSpace(6e,6o) Energy Results

Molecule	Formula	Electronic Ground State Energy (Eh)	Nuclear Repulsion Energy (Eh)	Total Ground State Energy (Eh)
Benzene	C_6H_6	-434.1365	203.1468	-230.9898
Naphthalene	$C_{10}H_8$	-840.9215	457.1060	-383.8155
Anthracene	$C_{14}H_{10}$	-1301.7013	765.0882	-536.6132
Pyrene	$C_{16}H_{10}$	-1577.6501	965.1896	-612.4604

Table 5: Runtime and memory for QDFT ActiveSpace(8e,6o)

Molecule	Formula	DFT (s)	Embedding (s)	VQE (s)	Total (s)	Peak RSS (MB)
Benzene	C_6H_6	17.21	5.40	1082.93	1110.22	1068.29
Naphthalene	$C_{10}H_8$	33.33	34.47	1155.19	1231.88	2350.23
Anthracene	$C_{14}H_{10}$	44.05	128.77	1170.04	1361.93	5745.11
Pyrene	$C_{16}H_{10}$	77.22	211.14	1225.66	1551.89	5220.77

Table 10: QDFT ActiveSpace(8e,6o) Energy Results

Molecule	Formula	Electronic Ground State Energy (Eh)	Nuclear Repulsion Energy (Eh)	Total Ground State Energy (Eh)
Benzene	C_6H_6	-434.1365	203.1468	-230.9898
Naphthalene	$C_{10}H_8$	-840.9063	457.1060	-383.8004
Anthracene	$C_{14}H_{10}$	-1301.6913	765.0882	-536.6031
Pyrene	$C_{16}H_{10}$	-1577.6430	965.1896	-612.4533

THANK YOU